5 **(a)** **(i)** The electric field strength at a point in an electric field is the electric force per unit positive charge on a small test charge at that point.

Note: Many stated 'force on a unit charge'.

(ii) The force acting on the charge $+q$ is $F = +qE$ in the direction of the field since it is a positive charge.

Work done $W = Fd = +qEd$

(iii) The potential difference V between two points is the work done per unit positive charge when a small test charge is moved between the two points.

$$V = \frac{W}{q} = \frac{+qEd}{q} = Ed$$

(b) **(i)** $n = \dfrac{8.6 \times 10^{-3}}{1.60 \times 10^{-19}} = 5.38 \times 10^{16}$ electrons s^{-1}

(ii) Gain in kinetic energy = Loss in electric potential energy

$$\frac{1}{2}mv^2 = eV$$

$$v = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2(1.6 \times 10^{-19})(60 \times 10^{3})}{9.11 \times 10^{-31}}} = 1.45 \times 10^{8} \text{ m s}^{-1}$$

(iii) The energy of each electron = 60 keV

The electrons arrive at a rate of 5.38×10^{16} s^{-1}

Power supplied $60(5.38 \times 10^{16}) = 3.23 \times 10^{21}$ eV s^{-1}

= 517 W

(c) Since X-ray production uses negligible fraction of the power, the 517 W has to be dissipated by the coolant.

$$\frac{dQ}{dt} = \frac{dm}{dt} c\Delta\theta$$

$$517 = \frac{dm}{dt}(3500)(30)$$

$$\frac{dm}{dt} = 4.92 \times 10^{-3} \text{ kg s}^{-1}$$

(d) **(i)** Extrapolating the field lines backwards, they meet at, and appear to originate from, the centre of the sphere. At distances $r \geq R$ (radius), the sphere's electric field is indistinguishable from that of a point charge at its centre.

Note: Many stated that the charge would be at the centre because the field lines are closer together near the centre. Comparatively few made a reference to the lines appearing to radiate from the centre.

(ii) $E = \dfrac{q}{4\pi\varepsilon_0 r^2} = \dfrac{0.060 \times 10^{-6}}{4\pi(8.85 \times 10^{-12})(0.10)^2} = 5.4 \times 10^4$ N C^{-1}

(iii) $V_{BC} = V_{AB}$

$$V_B - V_C = V_A - V_B$$

$$\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_B} - \frac{1}{r_C}\right) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_A} - \frac{1}{r_B}\right)$$

$$\frac{1}{r_B} - \frac{1}{r_C} = \frac{1}{r_A} - \frac{1}{r_B}$$

$$\frac{1}{0.50} - \frac{1}{r_C} = \frac{1}{0.40} - \frac{1}{0.50}$$

$$r_C = 0.67 \text{ m}$$

6 **(a)** Frequency is the number of complete oscillations per unit time, measured in hertz